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# GENERAL CHEMISTRY I

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Office hour: by appointment

Textbook: Chemistry: The Central Science, By Theodore E. Brown

# Expected Learning Outcomes

1. Understand the basic principles of general chemistry
2. Be able to explain basic, everyday chemical phenomena and apply your knowledge to solving real-world problems
3. Be able to explain why chemistry is essential and how it applies to everyday life

# 5 ways to fail this course

- Use communication devices in the exams
- Do not fully participate in the lab sections
- Take more than 3 absent days with or without the faculty's permission
- Fake lab data
- Plagiarize

What's your own definition  
of chemistry?

# GENERAL CHEMISTRY I

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## Chapter 1 Introduction: Matter and Measurement

# Contents

- 1-1 The Study of Chemistry
- 1-2 Classification of Matter
- 1-3 Properties of Matter
- 1-4 Units of Measurement: SI (Metric) System
- 1-5 Uncertainties in Scientific Measurements
- 1-6 Significant Figures

## Nutrition Facts

Serving Size 12 fl oz (360 mL)

Servings Per Container 6

Amount Per Serving

Calories 140

% Daily Value\*

Total Fat 0g 0%

Sodium 45mg 2%

Total Carbohydrate 39g 13%

Sugars 39g

Protein 0g

Not a significant source of calories from fat, saturated fat, trans fat, cholesterol, dietary fiber, vitamin A, vitamin C, calcium and iron.

\*Percent Daily Values are based on a 2,000 calorie diet.

CAR  
COL  
FLAV  
CAFF  
©20  
CON  
CALL  
WWW

Coke®



**Chemistry** is all around us

### Energy

Solar panels are composed of specially treated silicon.



### Biochemistry

The flash of the firefly results from a chemical reaction in the insect.



### Technology

LED's (light emitting diodes) are formed from elements such as gallium, arsenic and phosphorus.



### Medicine

Connectors and tubing for medical procedures such as intravenous injections are made from plastics highly resistant to chemical attack.



## Chemistry

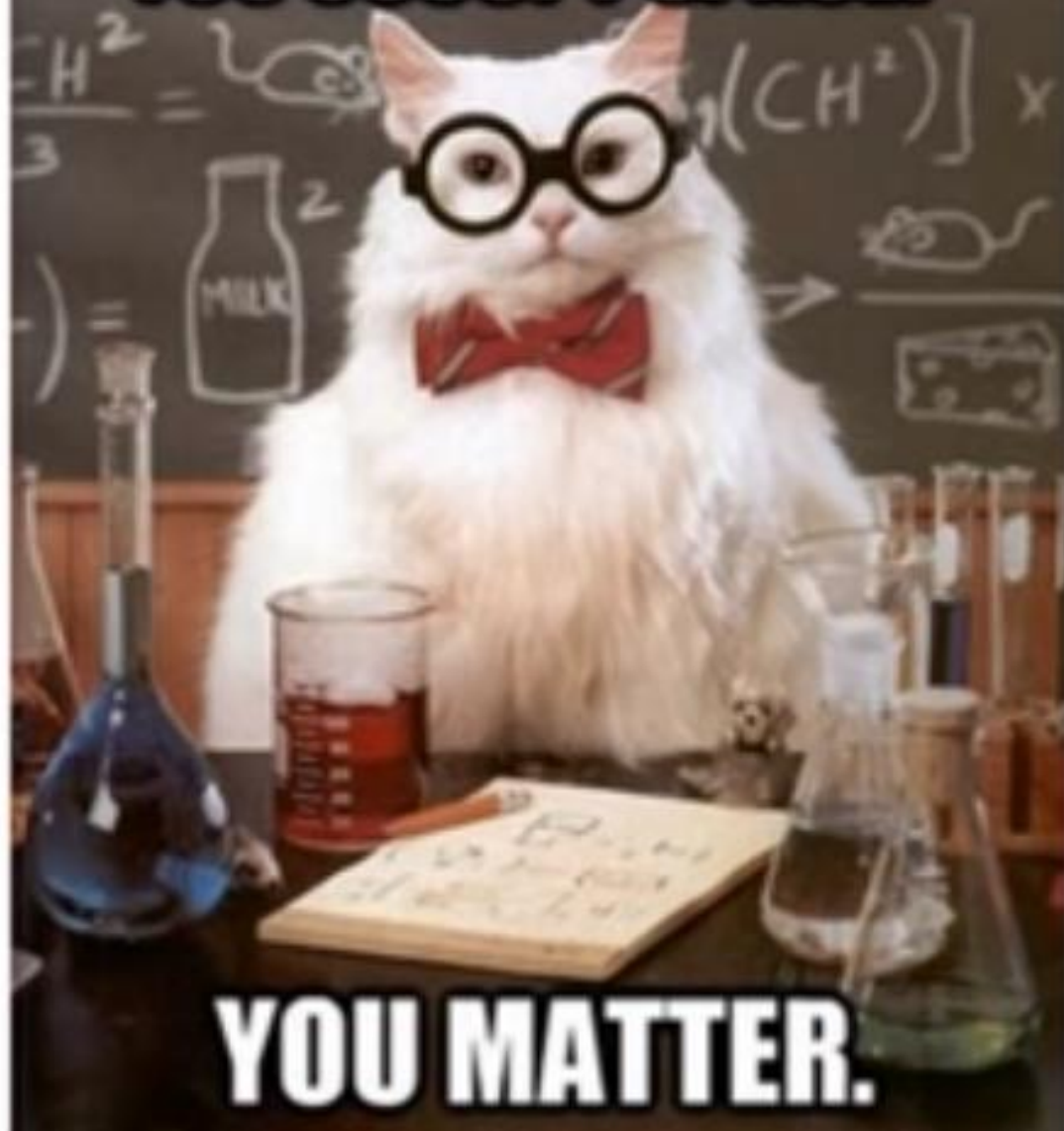
**Chemistry** is central to our understanding of the world around us

# 1-1 The Study of Chemistry



**Chemistry** is the study of matter  
and the changes that matter undergoes

**DO YOU HAVE MASS? DO  
YOU OCCUPY SPACE?**



**YOU MATTER.**

# The Periodic Table of the Elements

# The Periodic Table of the Elements

|          |                         |                       |                           |                              |                          |                           |                          |                        |                           |                             |                            |                            |                           |                          |                             |                            |                           |                            |
|----------|-------------------------|-----------------------|---------------------------|------------------------------|--------------------------|---------------------------|--------------------------|------------------------|---------------------------|-----------------------------|----------------------------|----------------------------|---------------------------|--------------------------|-----------------------------|----------------------------|---------------------------|----------------------------|
| group 1  |                         |                       |                           |                              |                          |                           |                          |                        |                           |                             |                            |                            |                           |                          |                             |                            | 18                        |                            |
| period 1 | 1<br>H<br>Hydrogen      |                       |                           |                              |                          |                           |                          |                        |                           |                             |                            |                            |                           |                          |                             |                            |                           | 2<br>He<br>Helium          |
| 2        | 3<br>Li<br>Lithium      | 4<br>Be<br>Beryllium  |                           |                              |                          |                           |                          |                        |                           |                             |                            |                            |                           |                          |                             |                            | 10<br>Ne<br>Neon          |                            |
| 3        | 11<br>Na<br>Sodium      | 12<br>Mg<br>Magnesium |                           |                              |                          |                           |                          |                        |                           |                             |                            |                            |                           |                          |                             |                            | 18<br>Ar<br>Argon         |                            |
| 4        | 19<br>K<br>Potassium    | 20<br>Ca<br>Calcium   | 21<br>Sc<br>Scandium      | 22<br>Ti<br>Titanium         | 23<br>V<br>Vanadium      | 24<br>Cr<br>Chromium      | 25<br>Mn<br>Manganese    | 26<br>Fe<br>Iron       | 27<br>Co<br>Cobalt        | 28<br>Ni<br>Nickel          | 29<br>Cu<br>Copper         | 30<br>Zn<br>Zinc           | 31<br>Ga<br>Gallium       | 32<br>Ge<br>Germanium    | 33<br>As<br>Arsenic         | 34<br>Se<br>Selenium       | 35<br>Br<br>Bromine       | 36<br>Kr<br>Krypton        |
| 5        | 37<br>Rb<br>Rubidium    | 38<br>Sr<br>Strontium | 39<br>Y<br>Yttrium        | 40<br>Zr<br>Zirconium        | 41<br>Nb<br>Niobium      | 42<br>Mo<br>Molybdenum    | (98)<br>Tc<br>Technetium | 44<br>Ru<br>Ruthenium  | 45<br>Rh<br>Rhodium       | 46<br>Pd<br>Palladium       | 47<br>Ag<br>Silver         | 48<br>Cd<br>Cadmium        | 49<br>In<br>Indium        | 50<br>Sn<br>Tin          | 51<br>Sb<br>Antimony        | 52<br>Te<br>Tellurium      | 53<br>I<br>Iodine         | 54<br>Xe<br>Xenon          |
| 6        | 55<br>Cs<br>Cesium      | 56<br>Ba<br>Barium    | 57<br>La<br>Lanthanum     | 58<br>Ce<br>Cerium           | 59<br>Pr<br>Praseodymium | 60<br>Nd<br>Neodymium     | 61<br>Pm<br>Promethium   | 62<br>Sm<br>Samarium   | 63<br>Eu<br>Europium      | 64<br>Gd<br>Gadolinium      | 65<br>Tb<br>Terbium        | 66<br>Dy<br>Dysprosium     | 67<br>Ho<br>Holmium       | 68<br>Er<br>Erbium       | 69<br>Tm<br>Thulium         | 70<br>Yb<br>Ytterbium      | 71<br>Lu<br>Lutetium      | 72<br>Hf<br>Hafnium        |
| 7        | 87<br>Fr<br>Francium    | 88<br>Ra<br>Radium    | 89<br>Ac<br>Actinium      | 90<br>Th<br>Thorium          | 91<br>Pa<br>Protactinium | 92<br>U<br>Uranium        | 93<br>Np<br>Neptunium    | 94<br>Pu<br>Plutonium  | 95<br>Am<br>Americium     | 96<br>Cm<br>Curium          | 97<br>Bk<br>Berkelium      | 98<br>Cf<br>Californium    | 99<br>Es<br>Einsteinium   | 100<br>Fm<br>Fermium     | 101<br>Md<br>Mendelevium    | 102<br>No<br>Nobelium      | 103<br>Lr<br>Lawrencium   | 104<br>Rf<br>Rutherfordium |
|          | (223)<br>Fr<br>Francium | (226)<br>Ra<br>Radium | (262)<br>Lr<br>Lawrencium | (261)<br>Rf<br>Rutherfordium | (262)<br>Db<br>Dubnium   | (266)<br>Sg<br>Seaborgium | (264)<br>Bh<br>Bohrium   | (277)<br>Hs<br>Hassium | (268)<br>Mt<br>Meitnerium | (271)<br>Ds<br>Darmstadtium | (272)<br>Rg<br>Roentgenium | (285)<br>Cn<br>Copernicium | (284)<br>Uut<br>Ununtrium | (289)<br>Fl<br>Flerovium | (288)<br>Uup<br>Ununpentium | (292)<br>Lv<br>Livermorium | 117<br>Uus<br>Ununseptium | (294)<br>Uuo<br>Ununoctium |

atomic mass  
or most stable mass number

1st ionization energy  
in kJ/mol

chemical symbol

name

electron configuration

55.845

26

Fe

Iron

[Ar] 3d<sup>6</sup> 4s<sup>2</sup>

atomic number

electronegativity

oxidation states  
most common are bold

alkali metals

alkaline metals

other metals

transition metals

lanthanoids

actinoids

metalloids

nonmetals

halogens

noble gases

unknown elements

radioactive elements have  
masses in parentheses

atomic mass  
or most stable mass number  
1st ionization energy  
in kJ/mol

chemical symbol

name

electron configuration

atomic number

electronegativity

oxidation states  
most common are bold

alkali metals

alkaline metals

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transition metals

lanthanoids

actinoids

metalloids

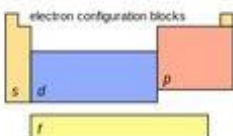
nonmetals

halogens

noble gases

unknown elements

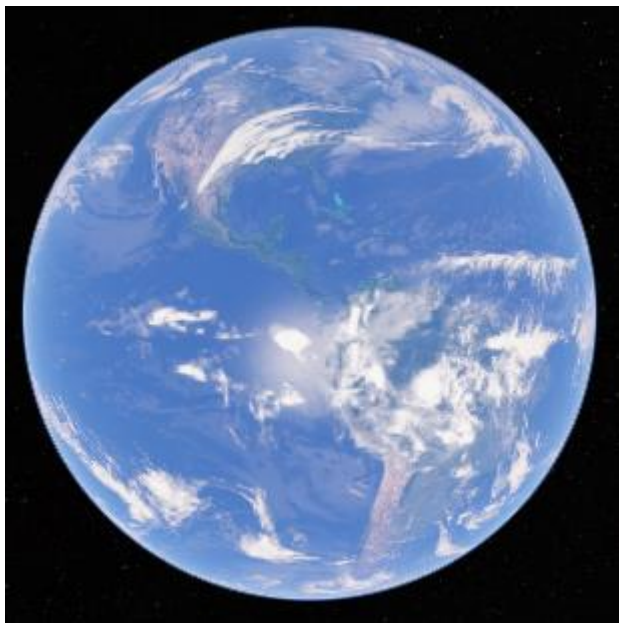
radioactive elements have  
masses in parentheses



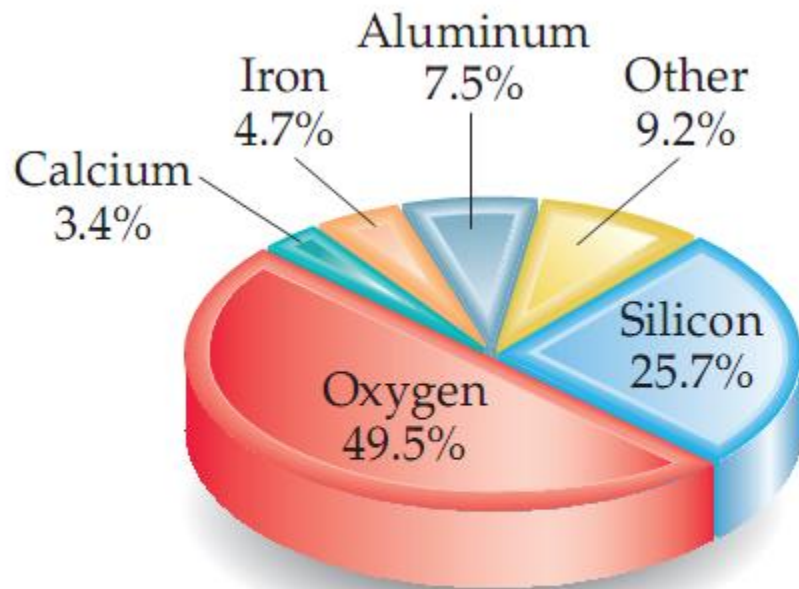
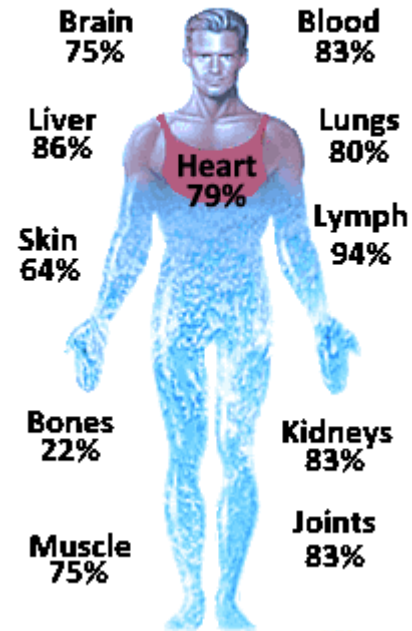
notes

- as of yet, elements 113, 115, 117 and 118 have no official name designated by the IUPAC.
- 1 kJ/mol = 96.485 eV.
- all elements are implied to have an oxidation state of zero.

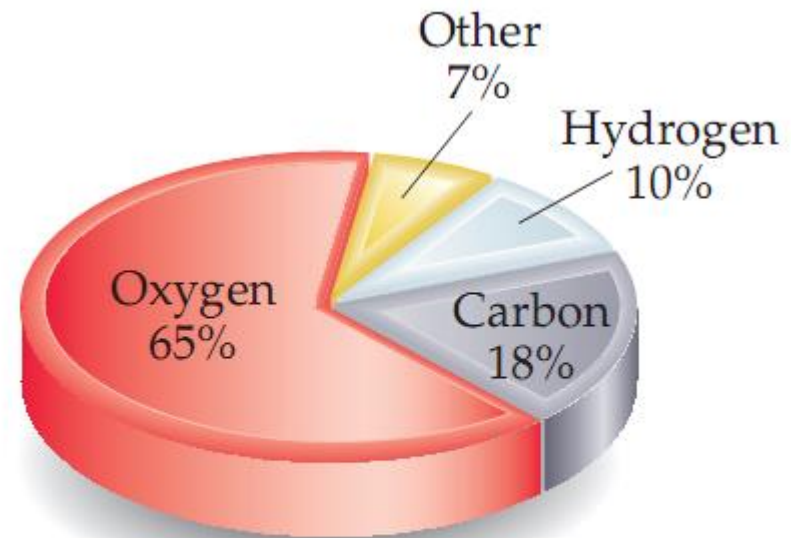
|                                |                              |                                   |                               |                              |                             |                              |                               |                              |                                |                               |                            |                                |                               |
|--------------------------------|------------------------------|-----------------------------------|-------------------------------|------------------------------|-----------------------------|------------------------------|-------------------------------|------------------------------|--------------------------------|-------------------------------|----------------------------|--------------------------------|-------------------------------|
| 138.9054 57<br>La<br>Lanthanum | 140.116 58<br>Ce<br>Cerium   | 140.9076 59<br>Pr<br>Praseodymium | 144.242 60<br>Nd<br>Neodymium | (145) 61<br>Pm<br>Promethium | 150.36 62<br>Sm<br>Samarium | 151.964 63<br>Eu<br>Europium | 157.25 64<br>Gd<br>Gadolinium | 158.9253 65<br>Tb<br>Terbium | 162.500 66<br>Dy<br>Dysprosium | 164.9303 67<br>Ho<br>Holmium  | 167.259 68<br>Er<br>Erbium | 168.9342 69<br>Tm<br>Thulium   | 173.054 70<br>Yb<br>Ytterbium |
| (227) 89<br>Ac<br>Actinium     | 232.0380 90<br>Th<br>Thorium | 231.0358 91<br>Pa<br>Protactinium | 238.0289 92<br>U<br>Uranium   | (237) 93<br>Np<br>Neptunium  | (244) 94<br>Pu<br>Plutonium | (243) 95<br>Am<br>Americium  | (247) 96<br>Cm<br>Curium      | (247) 97<br>Bk<br>Berkelium  | (251) 98<br>Cf<br>Californium  | (252) 99<br>Es<br>Einsteinium | (257) 100<br>Fm<br>Fermium | (258) 101<br>Md<br>Mendelevium | (259) 102<br>No<br>Nobelium   |



## Our Body is 60% ~70% Water



Earth's crust



Human body

# Atoms

**Atoms:** the almost infinitesimally small building blocks of matter

## The Scale of the Universe 2

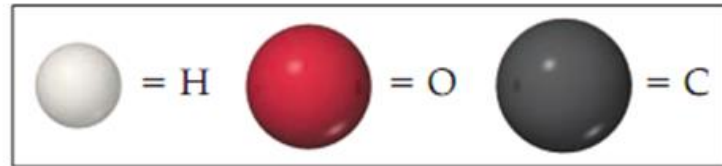
# Molecules and compounds

**Molecule:** two or more atoms are joined in specific shapes

Homonuclear molecule



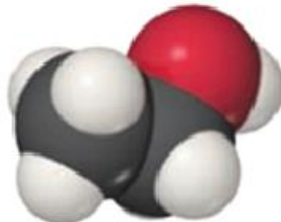
Oxygen



Compound



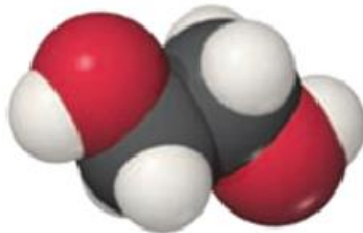
Water



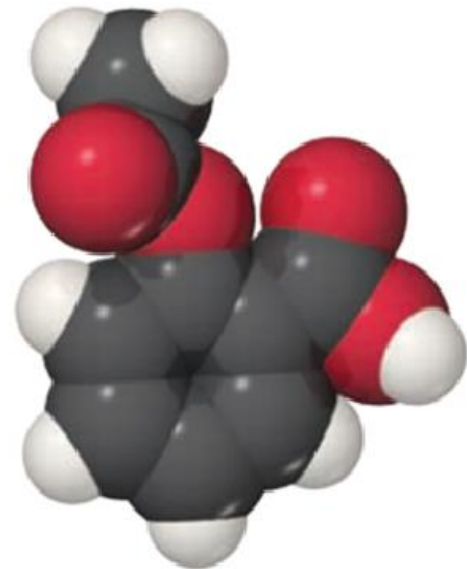
Ethanol



Carbon dioxide

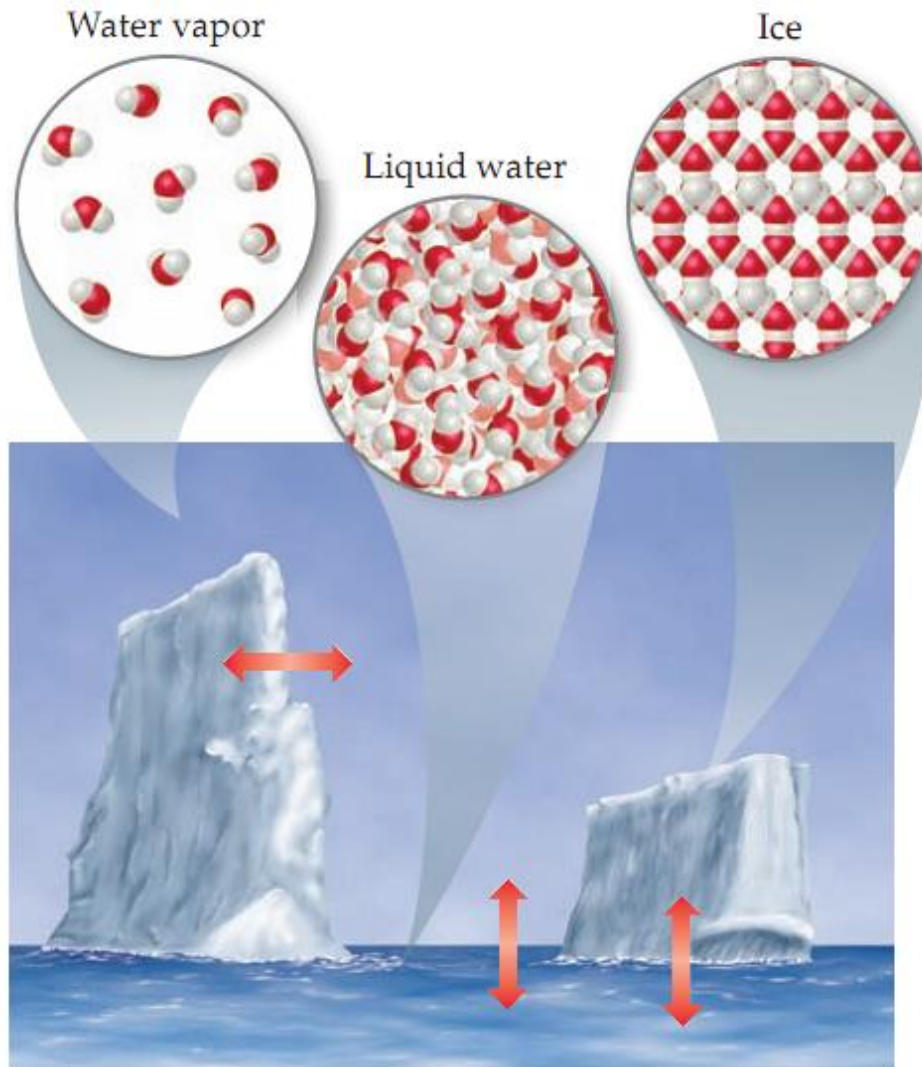


Ethylene glycol



Aspirin

# 1-2 Classification of Matter

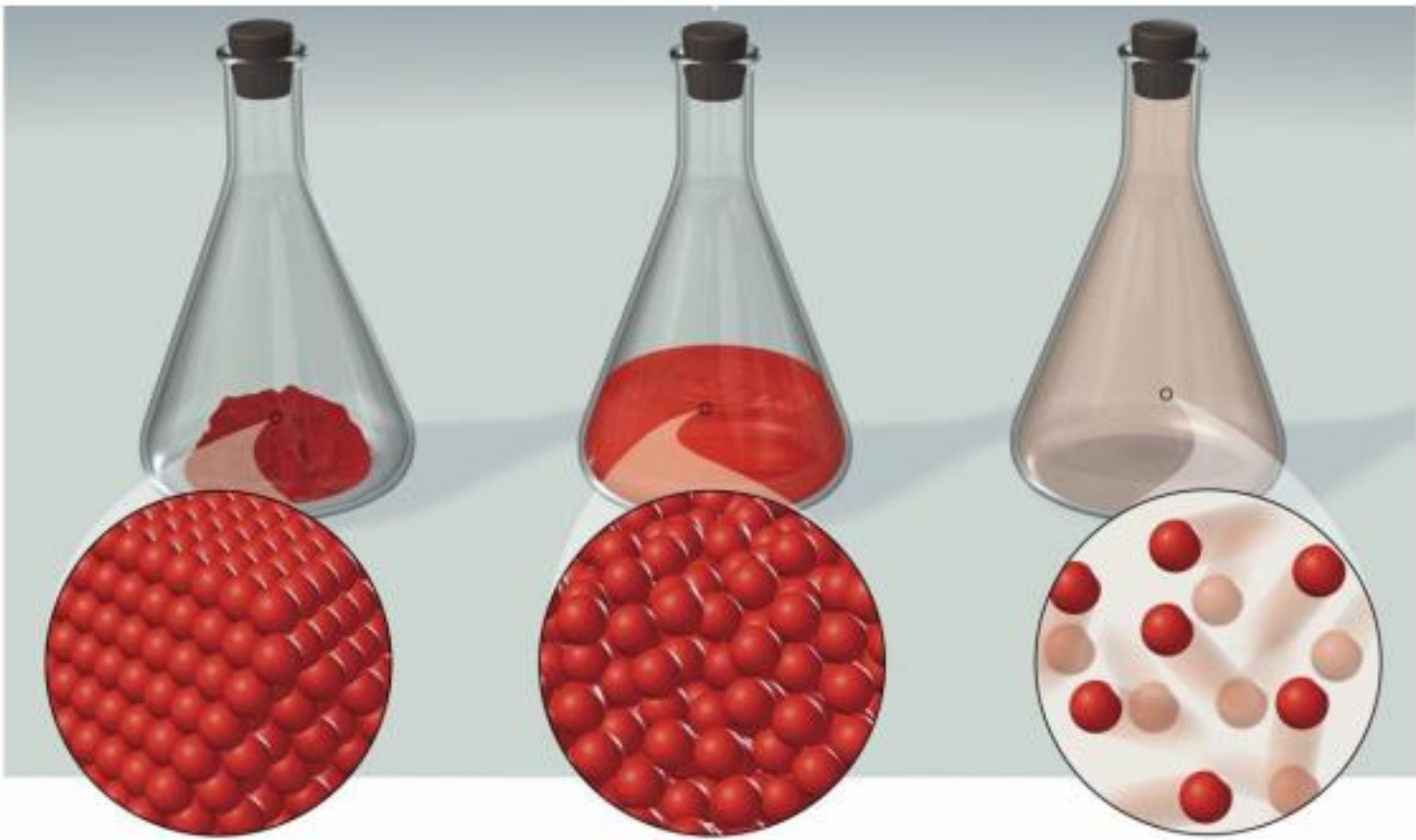


In which state of matter (gas, liquid or solid) are individual molecular furthest apart?

In which one they are the second furthest apart?

In which state are they closest together?

## Physical States of Matter



## **Solids**

have a fixed shape  
and volume

**not** compressible

## **Liquids**

conform to the shape  
of a container, but not  
the volume

**not** compressible

## **Gases**

take up the shape  
and volume of a  
container

**compressible**

# Pure substances

**Pure substances:** matter that has distinct properties and a composition that does not vary from sample to sample



Diamond (carbon, C)

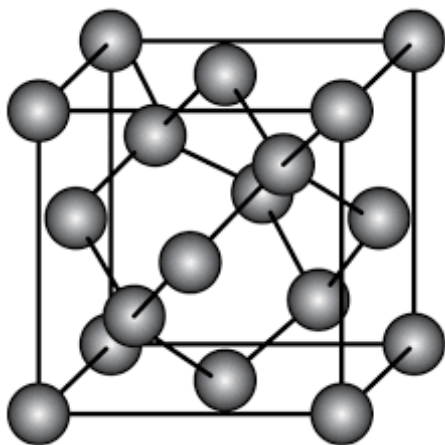
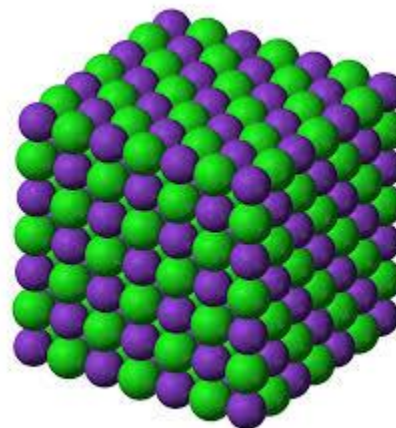


Table salt (sodium chloride, NaCl)



# Mixtures

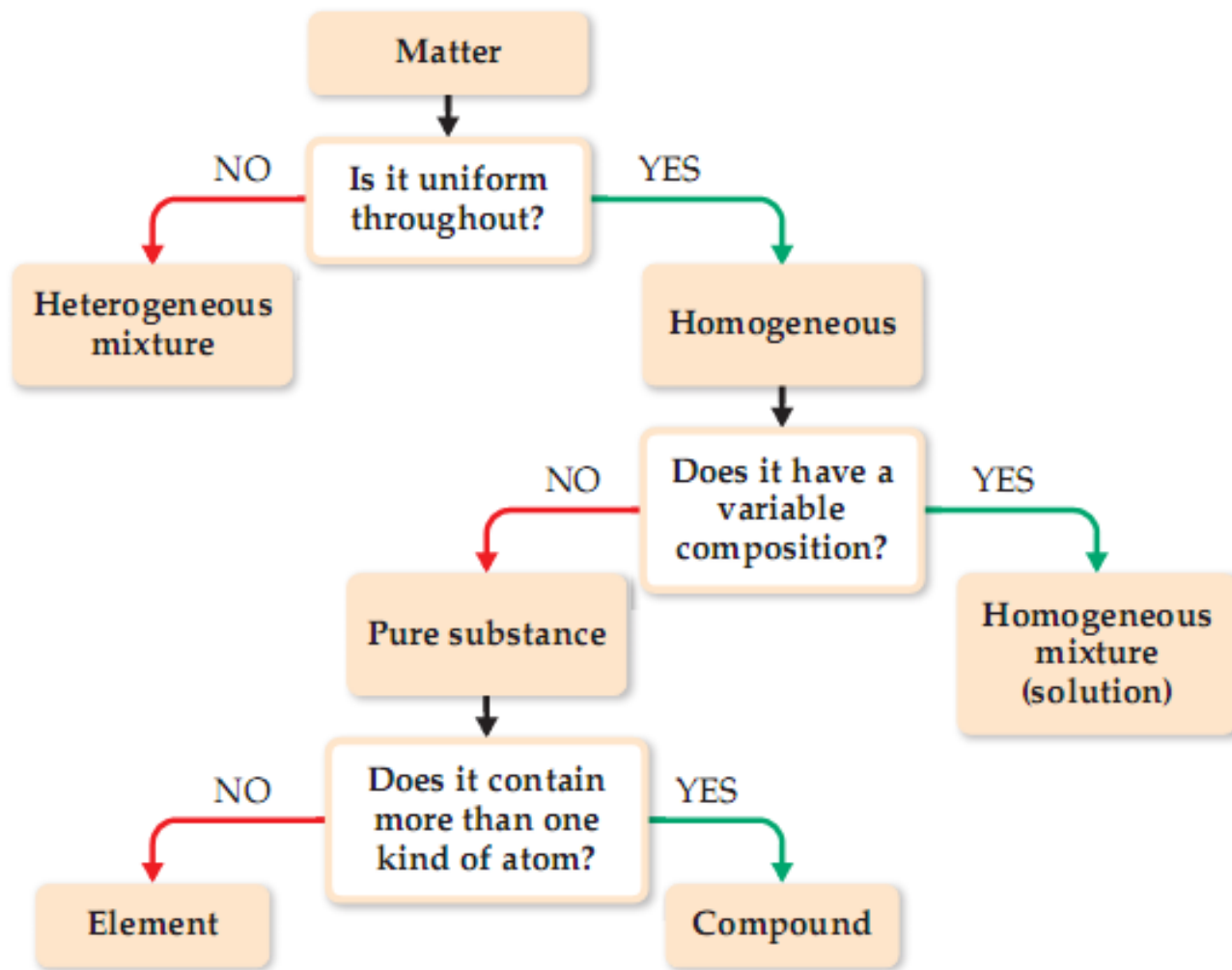
**Mixture:** collection of two or more pure substances



rocks  
a heterogeneous mixtures

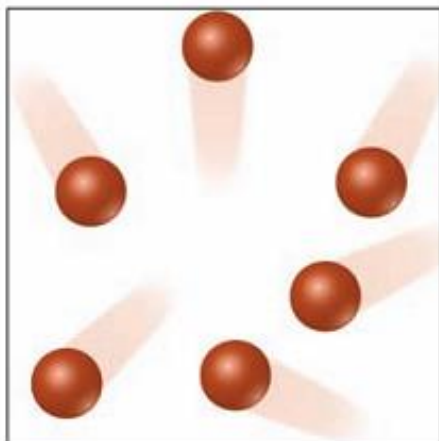


copper(II) sulfate aqueous solution  
a homogeneous mixtures

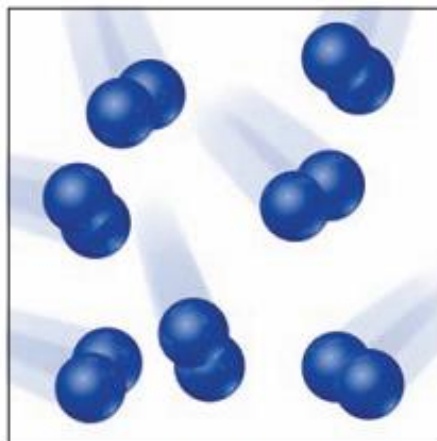


- **Matter** is made of atoms.
- **Atoms** are differentiated as **elements** (118 elements known).
- **Compounds** are comprised of two or more elements.
- **Molecules** are the smallest units of compounds.

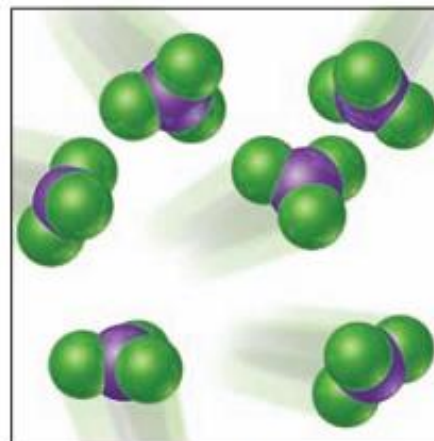
# How do the molecules of a compound differ from the molecules of an element?



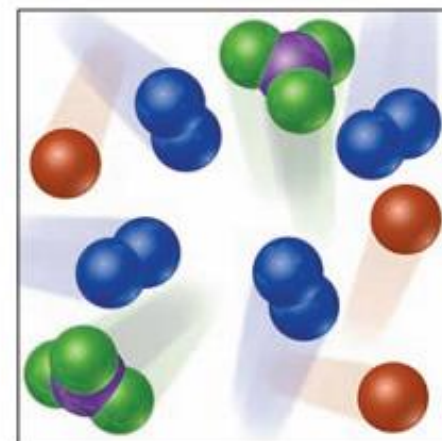
(a) Atoms of an element



(b) Molecules of an element



(c) Molecules of a compound

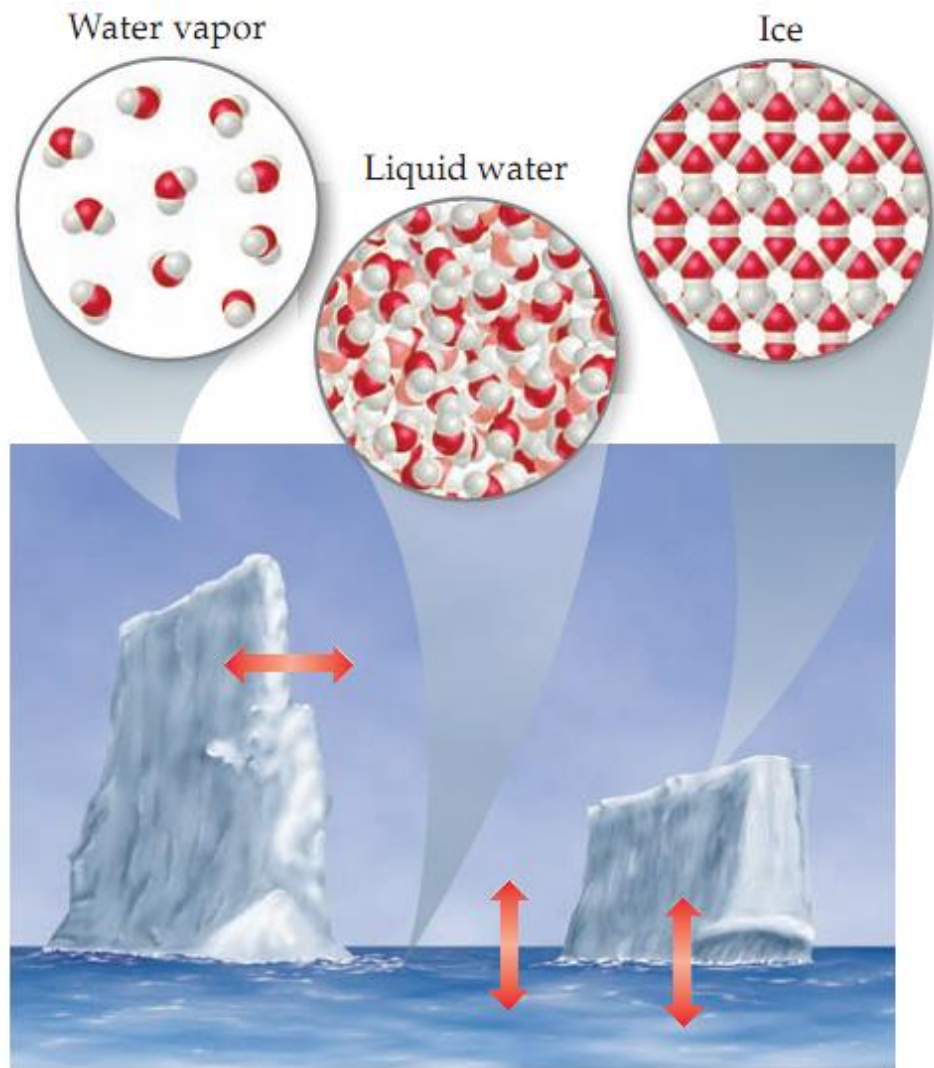


(d) Mixture of elements and a compound

Only one kind of atom is in any element.

Compounds must have at least two kinds of atoms.

# 1-3 Properties of Matter

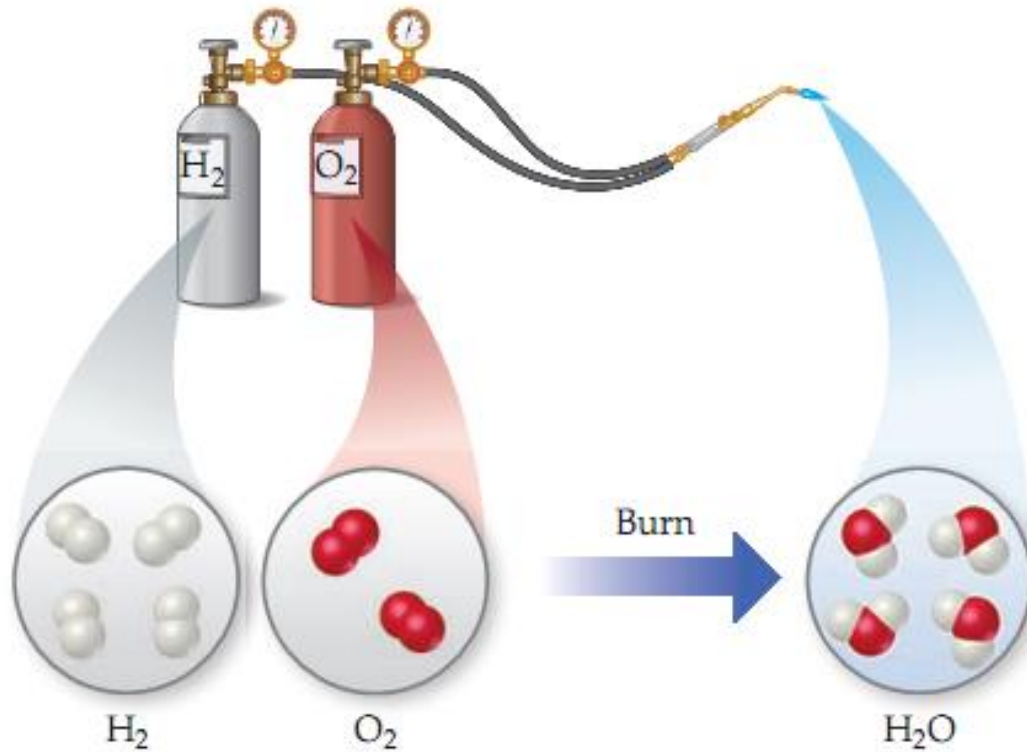


Physical  
change

**Physical appearance changes but chemical composition does not**

# Chemical change

**Chemical change** (also called a **chemical reaction**): a substance is transformed into a chemically different substance



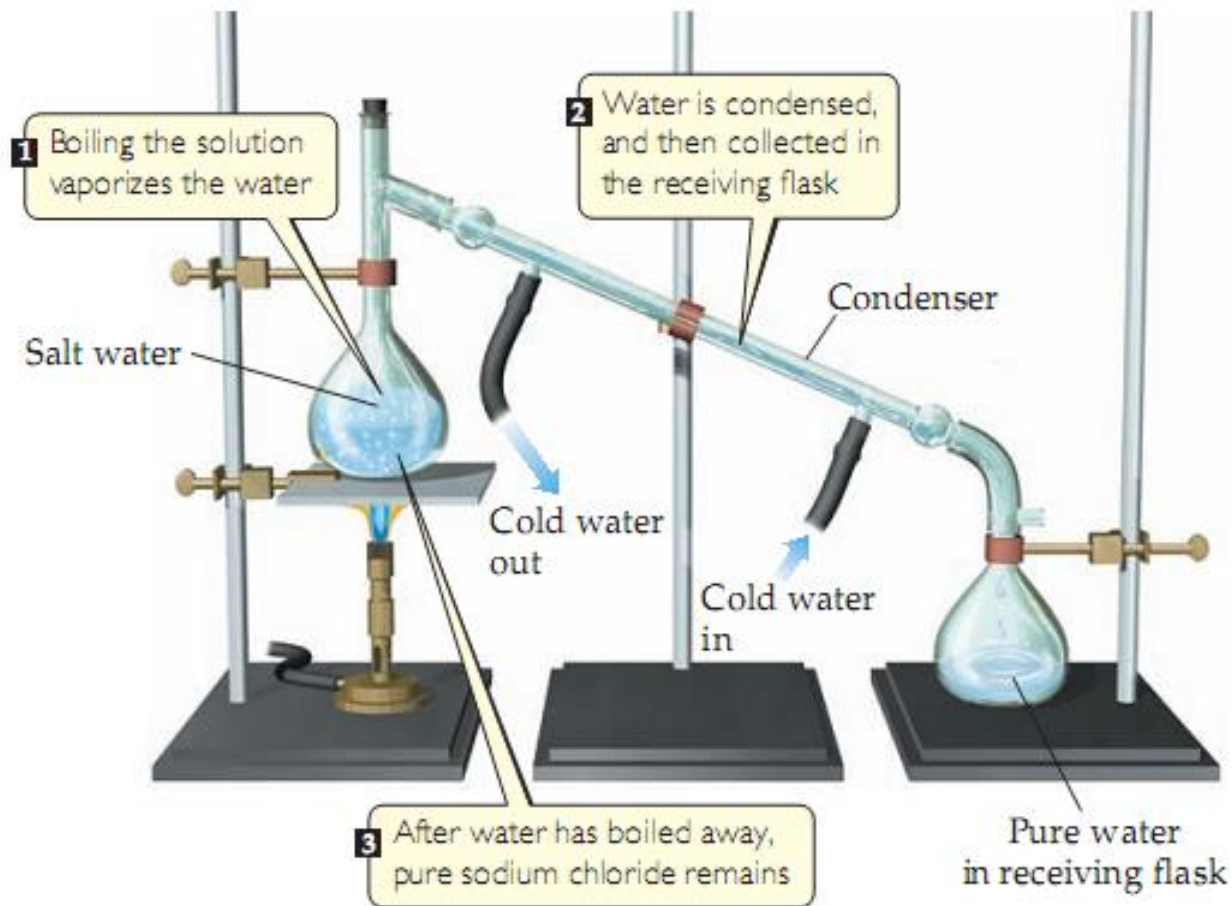
**Chemical change – formation of a new compound (water)**



**Chemical change – decomposition of a compound (ammonium chromate)**

# Separation of Mixtures

**Physical change** : a substance changes its physical appearance but not its composition



**Separation of a mixture (salt water) by distillation**

# EXAMPLE

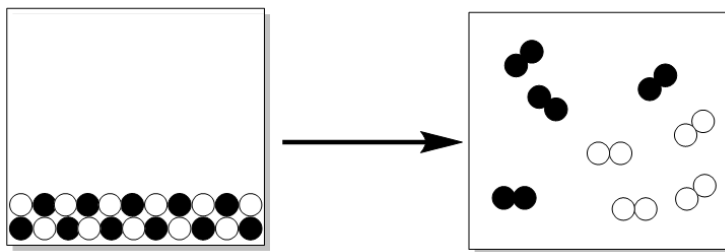
Which of the following represents a physical change?

- a. *Alkanes burn spontaneously.*
- b. *The sublimation of  $\text{CO}_2$ .*
- c.  $2 \text{H}_2 (\text{g}) + \text{O}_2 (\text{g}) \rightarrow 2 \text{H}_2\text{O} (\text{g})$
- d. *The rusting of a car.*

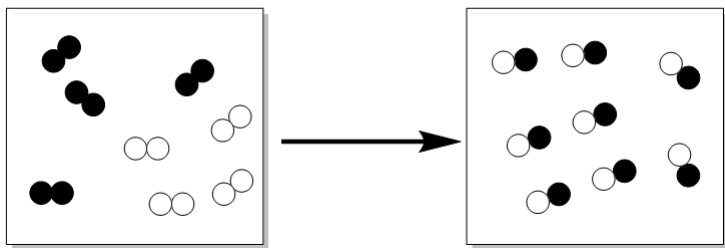
# EXAMPLE

Determine which of the following represents a physical change and a chemical change.

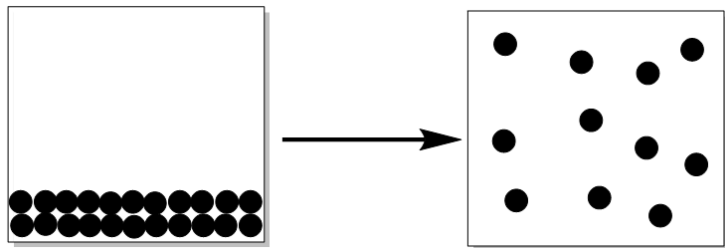
*a.*



*b.*

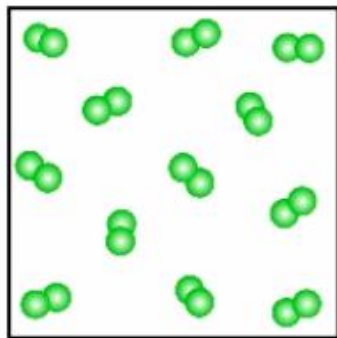


*c.*

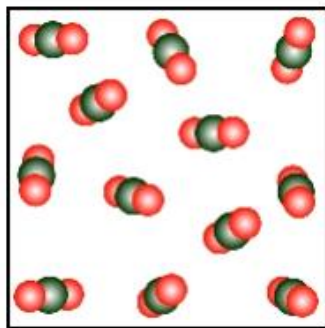


# EXAMPLE

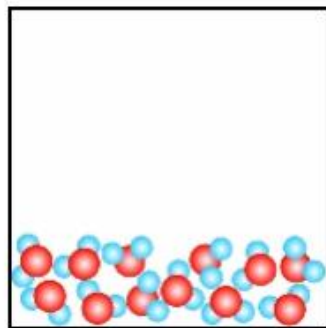
Answer each of the following questions based on the images provided below.



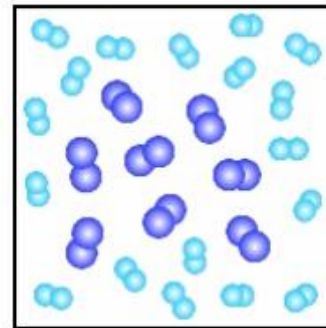
1



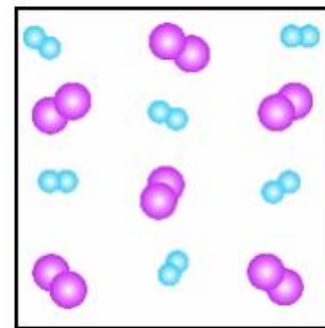
2



3



4



5

- a) Which images represents an elemental gas?
- b) Which images represents a heterogeneous mixture?
- c) Which images represents a liquid?

# EXAMPLE

Identify each of the following homogeneous or heterogeneous mixture

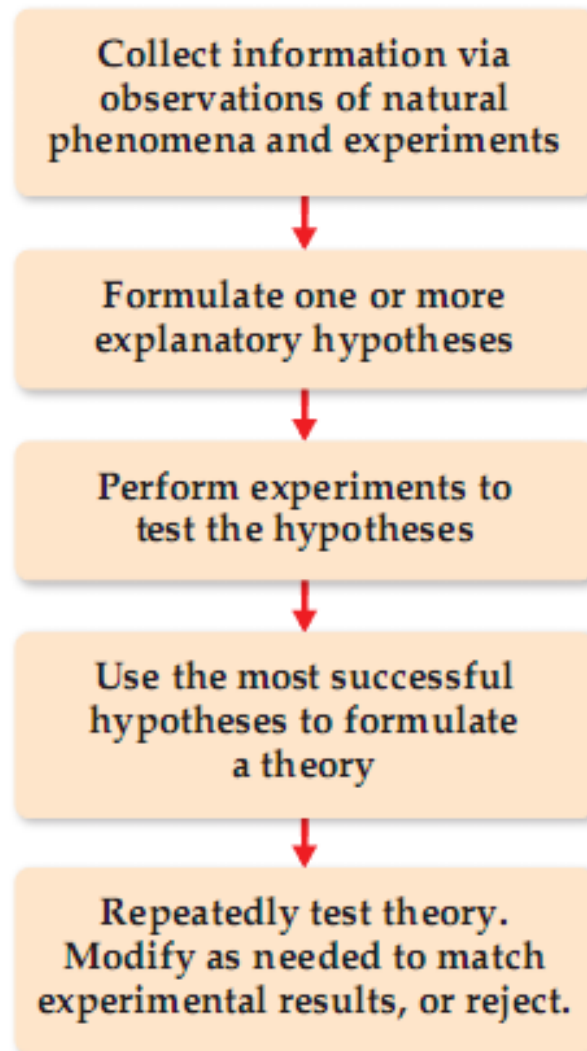
*a) Wood*

*b) Tap water*

*c) Dirt*

*d) Air*

# 1-4 The Measurement of Matter



## Scientific Method

**Table 1.4 SI Base Units**

| Physical Quantity   | Name of Unit | Abbreviation |
|---------------------|--------------|--------------|
| Mass                | Kilogram     | kg           |
| Length              | Meter        | m            |
| Time                | Second       | s or sec     |
| Temperature         | Kelvin       | K            |
| Amount of substance | Mole         | mol          |
| Electric current    | Ampere       | A or amp     |
| Luminous intensity  | Candela      | cd           |

**Table 1.5 Prefixes Used in the Metric System and with SI Units**

| Prefix | Abbreviation | Meaning    | Example                 |                                      |
|--------|--------------|------------|-------------------------|--------------------------------------|
| Peta   | P            | $10^{15}$  | 1 petawatt (PW)         | $= 1 \times 10^{15} \text{ watts}^a$ |
| Tera   | T            | $10^{12}$  | 1 terawatt (TW)         | $= 1 \times 10^{12} \text{ watts}$   |
| Giga   | G            | $10^9$     | 1 gigawatt (GW)         | $= 1 \times 10^9 \text{ watts}$      |
| Mega   | M            | $10^6$     | 1 megawatt (MW)         | $= 1 \times 10^6 \text{ watts}$      |
| Kilo   | k            | $10^3$     | 1 kilowatt (kW)         | $= 1 \times 10^3 \text{ watts}$      |
| Deci   | d            | $10^{-1}$  | 1 deciwatt (dW)         | $= 1 \times 10^{-1} \text{ watt}$    |
| Centi  | c            | $10^{-2}$  | 1 centiwatt (cW)        | $= 1 \times 10^{-2} \text{ watt}$    |
| Milli  | m            | $10^{-3}$  | 1 milliwatt (mW)        | $= 1 \times 10^{-3} \text{ watt}$    |
| Micro  | $\mu^b$      | $10^{-6}$  | 1 microwatt ( $\mu W$ ) | $= 1 \times 10^{-6} \text{ watt}$    |
| Nano   | n            | $10^{-9}$  | 1 nanowatt (nW)         | $= 1 \times 10^{-9} \text{ watt}$    |
| Pico   | p            | $10^{-12}$ | 1 picowatt (pW)         | $= 1 \times 10^{-12} \text{ watt}$   |
| Femto  | f            | $10^{-15}$ | 1 femtowatt (fW)        | $= 1 \times 10^{-15} \text{ watt}$   |
| Atto   | a            | $10^{-18}$ | 1 attowatt (aW)         | $= 1 \times 10^{-18} \text{ watt}$   |
| Zepto  | z            | $10^{-21}$ | 1 zeptowatt (zW)        | $= 1 \times 10^{-21} \text{ watt}$   |

# Scientific Notation

Scientific notation is used to turn small or large, inconvenient numbers into manageable ones

The diagram shows the scientific notation  $6.88 \times 10^{-12}$ . A blue arrow points from the word "digits" to the number 6.88. A purple arrow points from the words "power of ten" to the  $10^{-12}$  part.

$$6.88 \times 10^{-12}$$

digits

power of ten

the decimal point placed after the first digit

Six point eighty eight times ten to the minus twelfth

# EXAMPLE

Convert the following numbers into scientific notation

*a. 377,000*

*b. 0.000101*

*c. 707.82*

*d.  $161.00 \times 10^7$*

# Mass

- Mass is the **quantity** of matter in an object.
- Weight is the force of gravity on an object

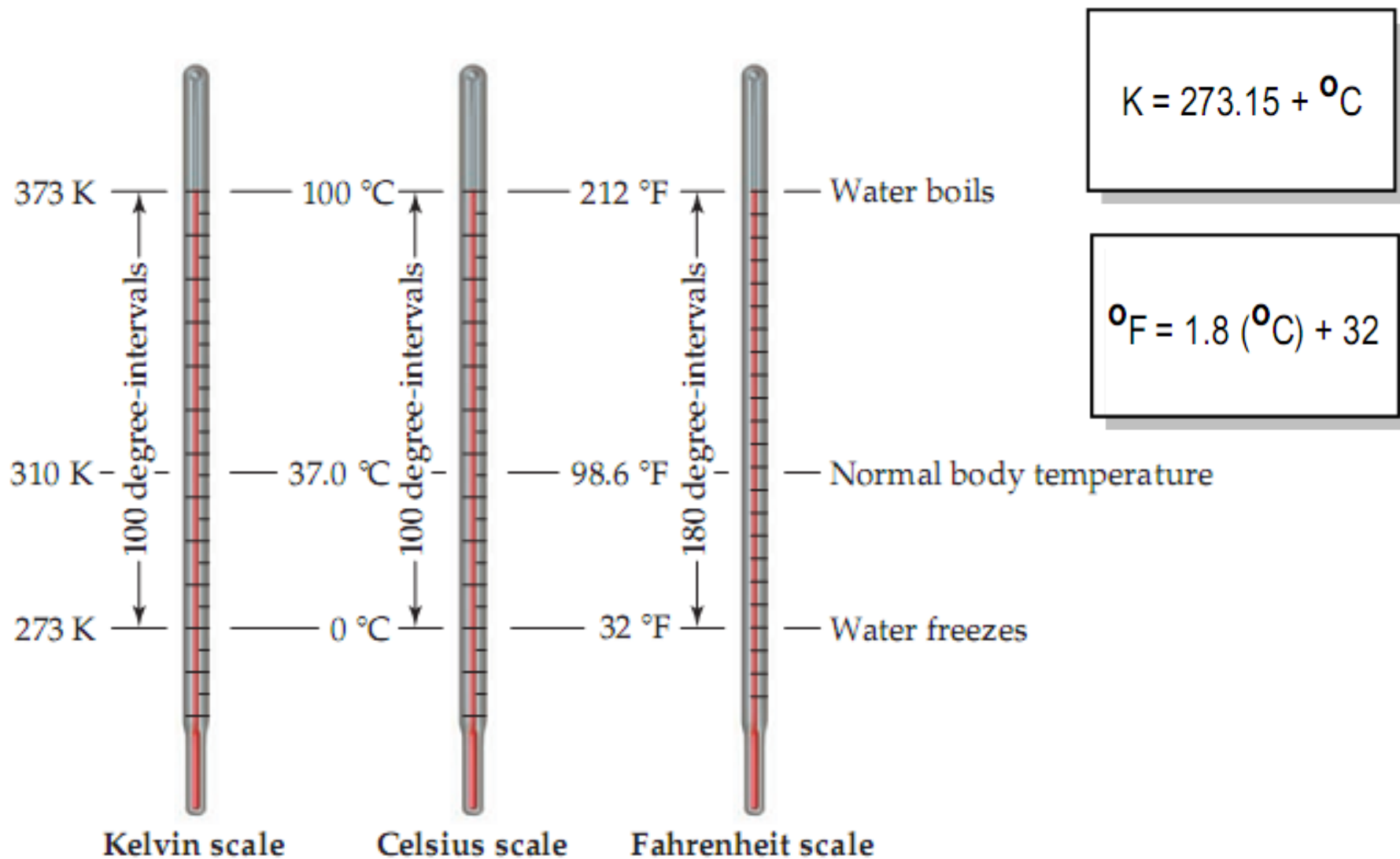
$$W \propto m$$

$$W = g \cdot m$$



# Temperature

Temperature is a measure of the hotness or coldness of an object

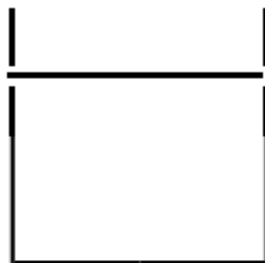


- **Temperature** is a measure of thermal energy in a substance, which is independent on the amount of matter.
  - This is an intensive property
  - Other examples of intensive properties: boiling point, melting point, color,...
- **Heat** is a form of thermal energy, which is dependent on the amount of matter.
  - This is an extensive property
  - Other examples of extensive properties: mole, mass, volume,...

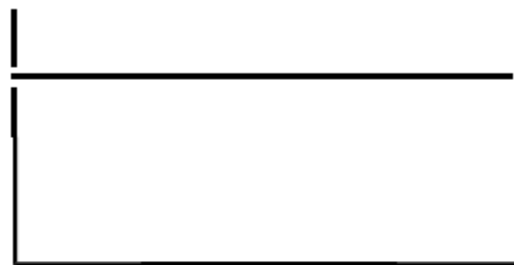
# EXAMPLE

Which of the following has the greater amount of heat?

a.

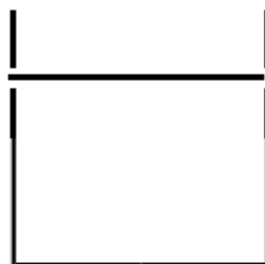


**30 g H<sub>2</sub>O at 50°C**

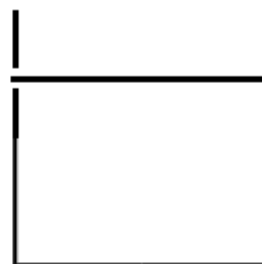


**300 g H<sub>2</sub>O at 50°C**

b.

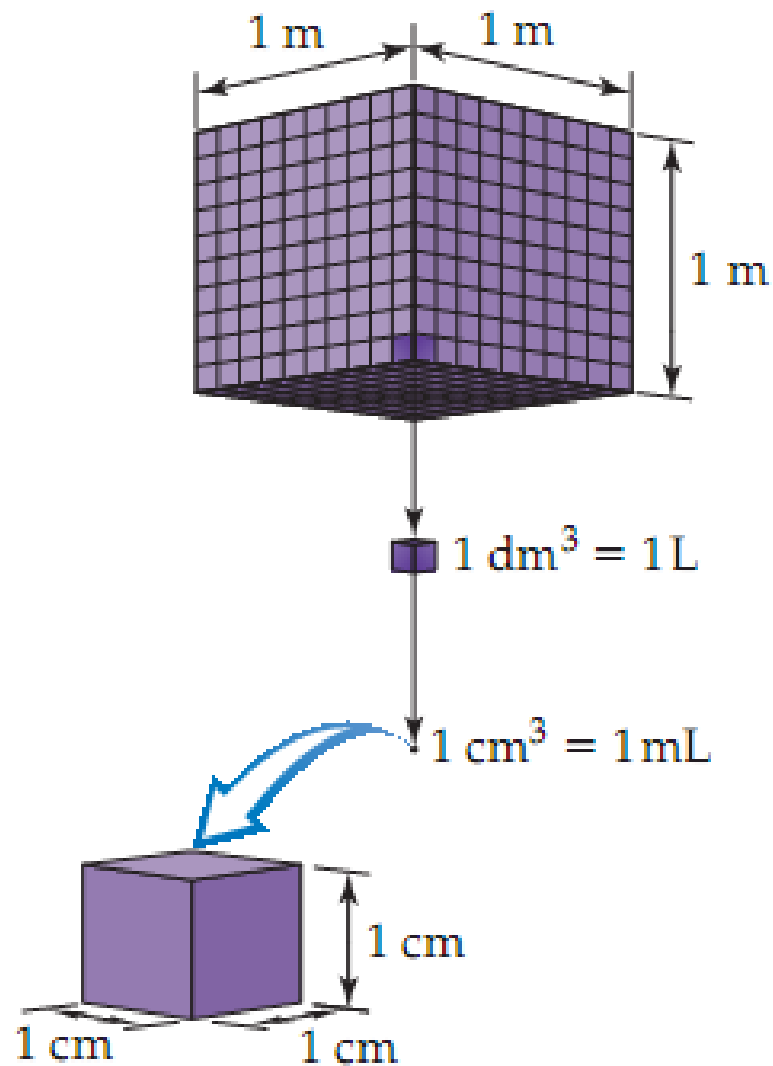


**30 g H<sub>2</sub>O at 50°C**



**30 g H<sub>2</sub>O at 100°C**

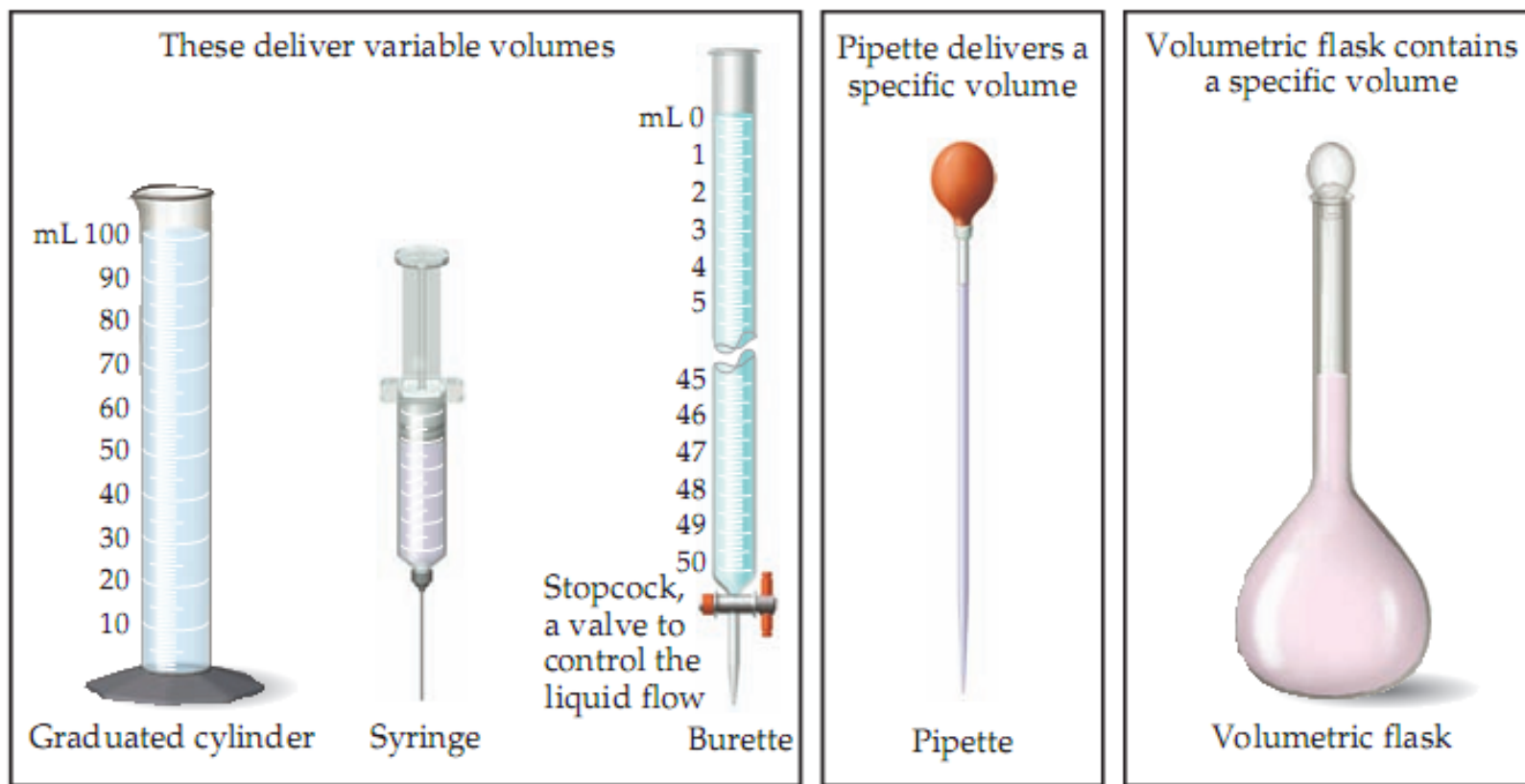
# Volume



# Density

Density is the amount of mass in a unit volume of a substance.

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$



**Common volumetric glassware for measuring volume of liquids**

# SI and non-SI Units Compared

1 Imperial qt  
1.136 L

1 US qt  
0.936 L

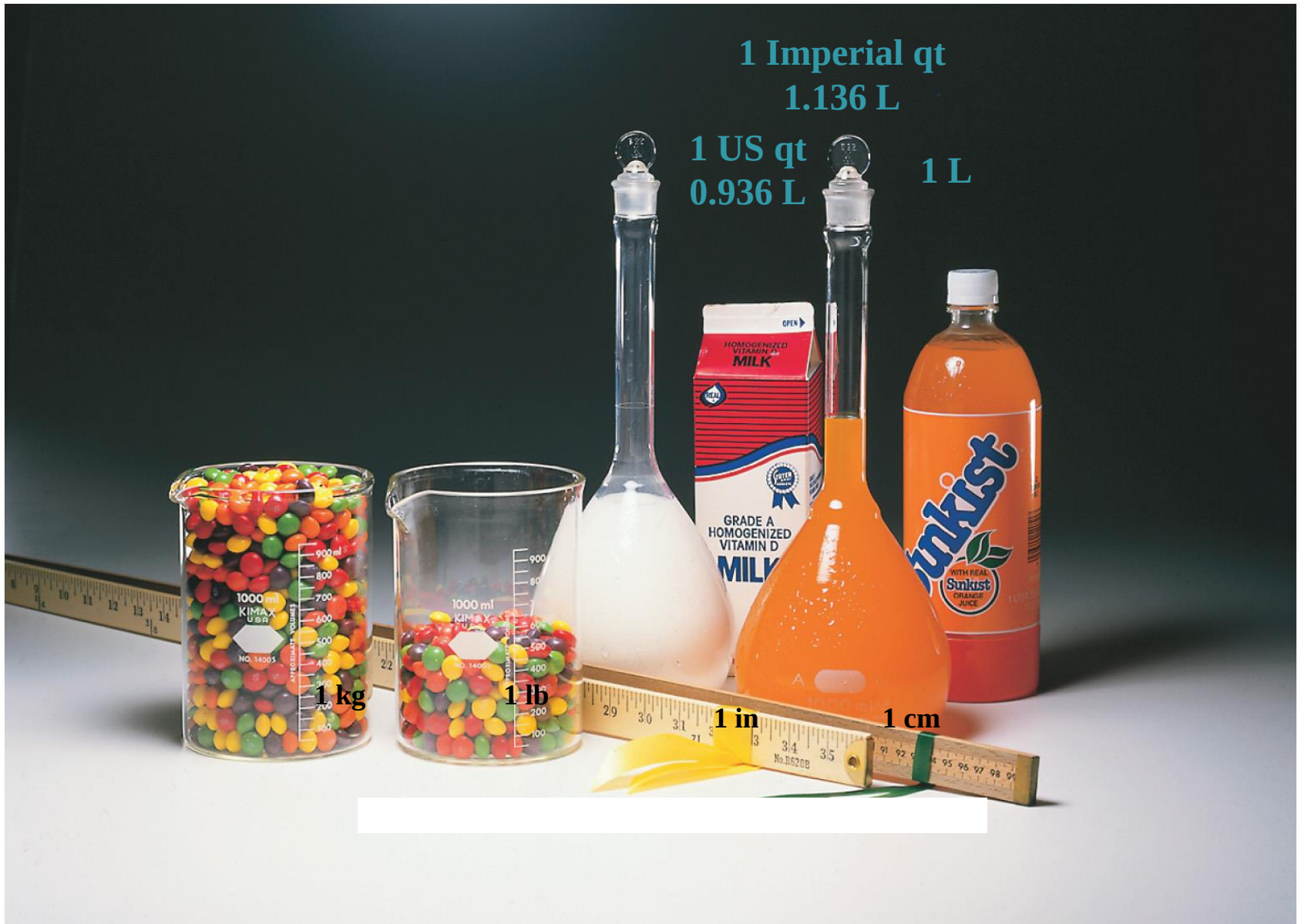
1 L

1 kg

1 lb

1 in

1 cm



# Units

## S.I. Units

|             |                                               |
|-------------|-----------------------------------------------|
| Length      | meter, m                                      |
| Mass        | Kilogram, kg                                  |
| Time        | second, s                                     |
| Temperature | Kelvin, K                                     |
| Quantity    | Mole, $6.022 \times 10^{23} \text{ mol}^{-1}$ |

## Other Common Units

|          |                                                 |
|----------|-------------------------------------------------|
| Length   | Angstrom, Å, $10^{-8} \text{ cm}$               |
| Volume   | Liter, L, $10^{-3} \text{ m}^3$                 |
| Energy   | Calorie, cal, 4.184 J                           |
| Pressure |                                                 |
|          | $1 \text{ Atm} = 1.064 \times 10^2 \text{ kPa}$ |
|          | $1 \text{ Atm} = 760 \text{ mm Hg}$             |

## Derived Quantities

|          |                                           |
|----------|-------------------------------------------|
| Force    | Newton, $\text{kg m s}^{-2}$              |
| Pressure | Pascal, $\text{kg m}^{-1} \text{ s}^{-2}$ |
| Energy   | Joule, $\text{kg m}^2 \text{ s}^{-2}$     |

# Density and Percent Composition

$$\delta = m/V$$

$$m = V\delta$$

$$V = m/\delta$$

$$\text{g/mL}$$

Mass and volume are **extensive** properties

Density is an **intensive** property

# Density in Conversion Pathways

What is the mass of a cube of osmium that is 1.25 inches on each side?

Have volume, need density = 22.48g/cm<sup>3</sup>

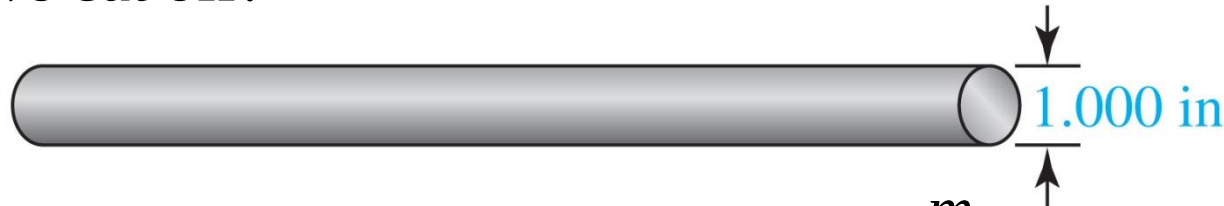
(converts in. to cm) (converts cm to cm<sup>3</sup>) (converts cm<sup>3</sup> to g osmium)

$$? \text{ g osmium} = \left[ 1.25 \text{ in.} \times \frac{2.54 \text{ cm}}{1 \text{ in.}} \right]^3 \times \frac{22.48 \text{ g osmium}}{1 \text{ cm}^3} = 719 \text{ g osmium}$$

# EXAMPLE

## *Relating Mass, Volume and Density*

The stainless steel in the solid cylindrical rod pictured below has a density of  $7.75 \text{ g/cm}^3$ . If we want a  $1.00 \text{ kg}$  mass of this rod, how long a section must we cut off?



Calculate the Volume needed

$$V = \frac{m}{d} = m \times \frac{1}{d}$$

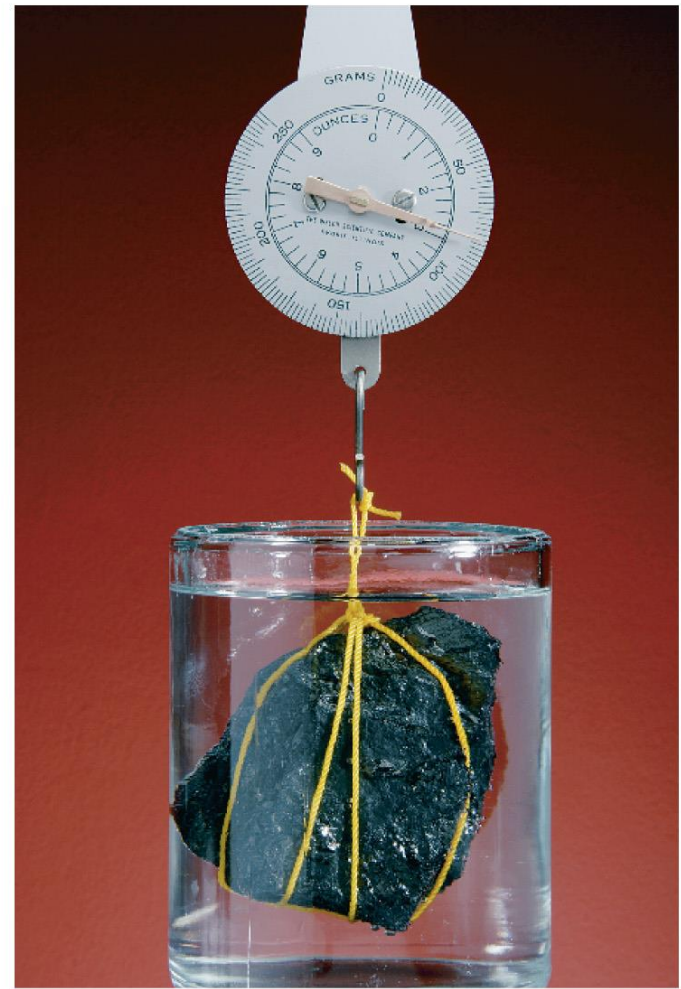
$$V = (1.00 \cancel{\text{kg}})(1000 \cancel{\text{g/kg}}) \times \frac{1 \text{ cm}^3}{7.75 \cancel{\text{g}}} = 129 \text{ cm}^3$$

Calculate the Length

$$V = h \pi r^2 \quad h = \frac{V}{\pi r^2}$$

$$h = \frac{V}{\pi r^2} = \frac{129 \text{ cm}^3}{(3.1415)\{(0.5)(1.000 \cancel{\text{in}})(2.54 \cancel{\text{cm/in}})\}^2} = 25.5 \text{ cm}$$

# Measuring Volume of an Irregular Object



# 1-5 Uncertainty in Measurement



Good accuracy  
Good precision

- Precision
  - Reproducibility of a measurement.



Poor accuracy  
Good precision

- Accuracy
  - How close to the real value.

- Systematic errors.
  - Constantly lower
  - Constantly higher



Poor accuracy  
Poor precision

- Random errors
  - Limitation in reading a scale.

# Precision



Reproducibility  $\pm 0.1$  g

Precision low

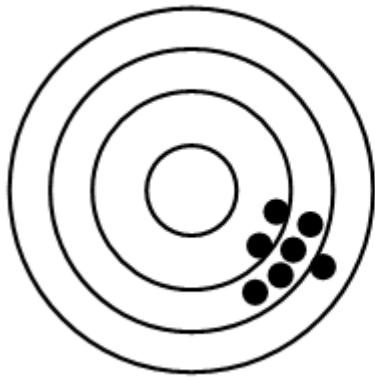


$\pm 0.0001$  g

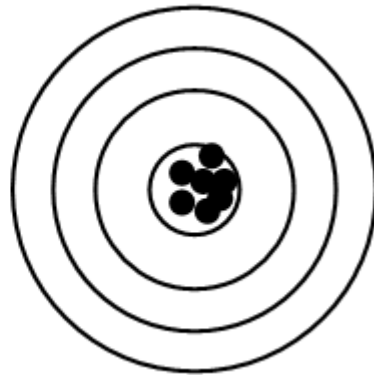
high

# EXAMPLE

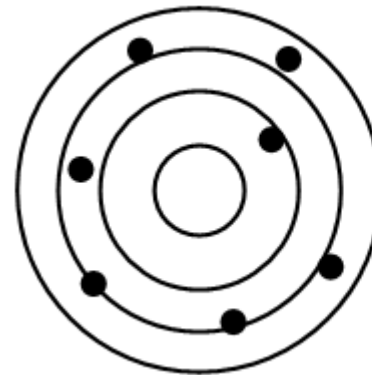
*Which of the 4 following images is not precise and not accurate?*



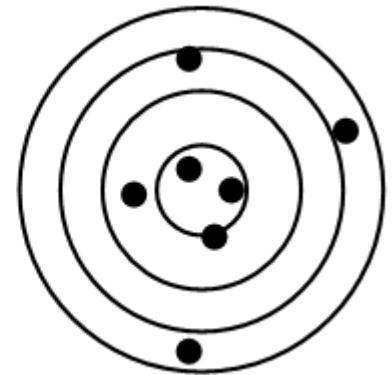
1



2



3



4

# EXAMPLE

A student must measure the weight of a sodium bicarbonate compound,  $\text{NaHCO}_3$ , and obtains the following measurements: 23.12 g, 23.08 g and 23.17g. If the true weight of the compound is 18.01 g what can be said about the student's results?

- a. *They are accurate and precise.*
- b. *They are accurate, but not precise.*
- c. *They are neither accurate or precise.*
- d. *They are not accurate, but precise.*

# Standard Deviation

The standard deviation measures **how close** data results are in relation to the mean, or average, value.

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$x_i$  = Observed value

$\bar{x}$  = Mean value

$n$  = Number of observations

# EXAMPLE

Calculate the standard deviation for the following results:

*a. 5.29, 5.35, and 5.31*

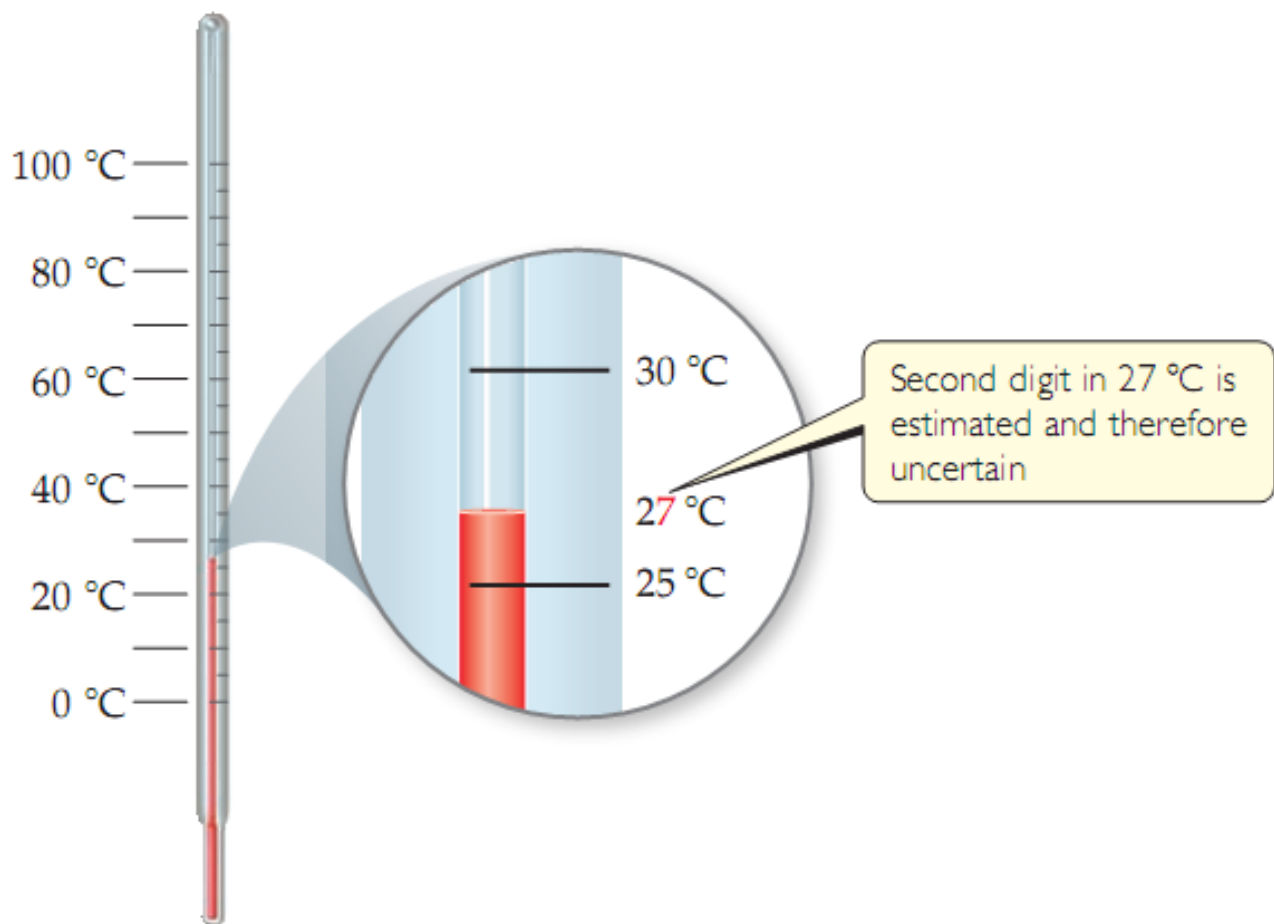
*b. 0.39, 0.061, and  $5.3 \times 10^{-2}$*

Now, answer this question:

**How tall are you?**

# Significant Figures

All digits of a measured quantity, including the uncertain one, are called significant figures



# Significant Figures

Not significant:

zero for  
“cosmetic”  
purpose

0

Not significant:

zeros used only  
to locate the  
decimal point

0

0

4

0

0

4

5

0

0

Significant:

all nonzero  
integers

Significant:

all zeros between  
nonzero numbers

Significant:

zeros at the end of  
a number to the right  
of decimal point

# EXAMPLE

How many sig figs does each number contain?

a.  $0.0000185\text{ m}$

b.  $749\text{ mol}$

c.  $17.3 \times 10^3\text{ mL}$

d.  $100\text{ min}$

e.  $0.0010050\text{ kg}$

f.  $1560\text{ mol}$

# Significant figures in calculation

For **addition and subtraction**, the result has the same number of decimal places as the measurement with the **fewest decimal places**

$$20.42 + 1.322 + 83.1 = 104.842$$

two decimal places

three decimal places

one decimal place

We report the result as **104.8** because 83.1 has only one decimal place

# Significant figures in calculation

For **multiplication and division**, the result contains the same number of significant figures as the measurement with the **fewest significant figures**

$$\text{Area} = (6.221 \text{ cm})(5.2 \text{ cm}) = 32.3492 \text{ cm}^2 \Rightarrow \text{round off to } 32 \text{ cm}^2$$

four significant figures

two significant figures

We report the result as **32 cm<sup>2</sup>** because 5.2 has two significant figures

Check it at <https://www.omnicalculator.com/math/sig-fig>

# EXAMPLE

Perform the following calculation to the right number of sig figs:

a.  $(3.16) \times (0.003027) \times (5.7 \times 10^{-3})$

$$0.000054522324 = 5.4522324 \times 10^{-5} = 5.5 \times 10^{-5}$$

b.  $2.628 \times 10^6$

$$6.281 \times 10^4$$

$$\underline{+ 0.827 \times 10^7}$$

$$10960180 = 1.0960180 \times 10^7 = 1.10 \times 10^7$$

# EXAMPLE

Perform the following calculation to the right number of sig figs:

a.  $(3.16) \times (0.003027) \times (5.7 \times 10^{-3})$

$= 5.5 \times 10^{-5}$

b.

|                       |                     |
|-----------------------|---------------------|
| $2.628 \times 10^6$   | $0.110 \times 10^8$ |
| $6.281 \times 10^4$   | $1.10 \times 10^7$  |
| $+ 0.827 \times 10^7$ |                     |

# Significant figures in calculation

When a calculation involves two or more steps and you write answers for intermediate steps, **retain at least one nonsignificant digit** for the intermediate answers.